QP CODE: 25900295

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU – UGP (HONOURS) REGULAR EXAMINATION October 2025

THIRD SEMESTER

Discipline Specific Core Course (DSC) –MG3DSCCND200

CLINICAL NUTRITION - I

(2024 ADMISSION ONWARDS)

Duration: 2 HoursMaximum Marks: 50

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Interest (I),Appreciation(Ap),and Skill(S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple ChoiceQuestions
Answer all questions
Each question carries 1 mark

1. Abdominal bloating
2. Cyclic infusion
3. Bacteria
4. Female anopheles mosquito
5. 30 - 39.9
6. Low fat

Part B
Short Answer
Answer any **2** questions.
Each question carries **2** marks.

7. **Recall TPN**
TPN – Total Parenteral Nutrition – ***0.5 Marks***
Provision of all nutrients through parenteral route, which is directly into the vein – ***1.5 marks***

8. **Explain why calorie requirement is increased in tuberculosis**
To maintain satisfactory weight – ***1 mark***
2500- 3000 Kcal is provided – ***1 mark***

9. **Discuss the role of sulphonylureas in controlling diabetes.**
- Increase body's production of insulin – **¼ mark**

- May provoke hypoglycemia and their use requires regular timing of meals and snacks - ¼ **mark**
- Tolbutamide is the safest of sulphonylureas. Effective action is 6 to 8 hrs and should be taken 2 to 3 times a day - ¼ **mark**
- Chloropropamide can be given as single dose of 100 to 375 mg/day. Toxic effect such as jaundice, drug rashes and blood dyscrasia (abnormal cells of blood) is seen in long term use. - ¼ **mark**

10. Discuss nasogastric tube feeding

- Feeding provided through nose into the stomach
- Recommended for patients with normal gastrointestinal functions
- Soft polyurethane tubes are used
- Verification can be done through aspirating gastric contents - ½ **mark each**

Part C

Short Essay

Answer any **4** questions

Each question carries **5** marks.

11. Differentiate elemental diet and semi elemental diet

Elemental diet – also known as hydrolysed formulas.

- Recommended for patients with limited gastrointestinal function.
- Normal foods replaced by preparation of protein or amino acids, fats and carbohydrates in easily assimilable form
- More expensive than general purpose formula – ***2. 5 marks***

Semi elemental diet – they are peptide based formulas also contain free amino acids.

- Carhohydrates are usually hydrolysed.
- Fat content is low – some may contain MCT or SFA.
- Beneficial for patients with short bowel syndrome or limited GI function – ***2.5 marks***

12. Explain the nutritional requirements in influenza

Influenza – Short duration fever. Spreads from person to person by inhalation of virus- ***1 mark***

Nutritional requirements ***(4 marks for Nutritional requirements)***

Energy – 50% increase in calorie requirement due to increased temperature and tissue destruction

Protein – About 100gm is recommended

Carbohydrate – high CHO to replenish glycogen storage

Fat – Moderate fat – avoid fried foods

Minerals – Sufficient intake of sodium chloride to compensate electrolyte loss

Vitamins – Increased requirement of Vitamin A, C and B complex vitamins.

Fluid – liberal. 2.5 – 5 lit/day is recommended.

13. Explain the recommendations for exercise in pregnancy.

- Exercise optimises glycemic control
- Emphasis on regular exercise to avoid complications of glycemic level
- Glycemic control should be assessed with doctor at every visit to adjust lifestyle and behaviour modifications
- Avoid strenuous exercise to prevent hypoglycaemia and oxygen saturation levels
- Helps to maintain blood pressure and blood circulation – **1 mark for each point**

14. Explain bolus method of infusion and continuous method of infusion in enteral feeding

Bolus infusion - When patient is clinically stable, syringe bolus method is recommended.

- They are less expensive than pump or gravity feeding method.
- 60 ml syringe is used to infuse the formula.
- If bloating occurs weight for 10- 15 minutes.
- Usually the patient can tolerate 500ml at each feeding – ***2.5 marks***

Continuous infusion – appropriate for patients who do not tolerate large volume infusions.

- Recommended for patients with compromised gastrointestinal function.
- Feeding provided usually 18- 24 hours daily.
- Formulas with 300- 500mOsm/kg can be started at full strength.- ***2.5 marks***

15. Summarize the macronutrient deficiency in AIDS

- Mild weight loss and wasting, body cell mass depletion due to fat depletion.
- Depletion of total body water in patients with malabsorptive illness.
- PEM is common - decreased skin fold thickness and mid arm circumference, decreased iron binding capacity and hypoalbuminemia.
- Metabolic irregularities and uncontrolled opportunistic infections – ***5 marks***

16. Differentiate between type 1 and type 2 diabetes

Type 1	**Type 2**
• Insulin dependent • Occur in younger age • Little or no production of insulin due to destruction of pancreatic cells • Patient is underweight • Acidosis is common	• Non-insulin dependent • Adult onset • Insulin is produced, but action is impaired • Patient is overweight • Acidosis is infrequent

2.5 + 2.5 = 5 marks

Part D

Essay

Answer any 2 questions

Each question carries 10 marks

17. Explain the complications and monitoring in parenteral nutrition

Complications-

Catheter related complications – cardiac arrhythmia, air embolus, thrombosis, perforation and sepsis- ***2.5 marks***

Metabolic complications – hyperglycemia, osmotic diuresis, azotemia, electrolye disturbance, acid base change, fatty degeneration of liver. – ***2.5 marks***

Nutritional complications – loss of lean body weight, bone demineralization, vitamin deficiency and toxicity essential fatty acid deficiency, trace metal deficiency ***– 2.5 marks***

Monitoring- routine monitoring is essential to avoid the risk of complications. Monitor the signs of infection, chills, fever and tachycardia, sudden hyperglycemia and elevated WBC. – **2.5 marks**

18. Explain AIDS – pathology, transmission and clinical manifestations

Pathology – HIV invades the genetic core of CD+ or T helper lymphocyte cells leads to its depletion and results in immunodeficiency- ***1 mark***

Transmission – transmitted by body fluids. ***3 marks***

Prenatal transmission - **2marks**

Clinical manifestation

Acute HIV – it is 4-7 weeks immediately after primary infection. Fever, malaise, lymphadenopathy syndrome, pharyngitis, headache, myalgia may occur.- ***1 mark***

Asymptomatic HIV – Last for 10 years. Decrease in lean body mass, vitamin B12 deficiency. ***-1 mark***

Symptomatic HIV – fever, sweat, skin problems and fartigue. Decline in nutrient status and body composition – ***1 mark***

AIDS – last stage of HIV. A small proportion infected with HIV may die. – **1 mark**

19. Explain the principles of diet and dietary management in diabetes

- Principles of diet – Low calorie, low carbohydrate, adequate protein, low fat, high fibre diet - **1 mark**
- Energy – 25 kcal/kg body weight - **½ mark**

- Carbohydrate – 45-60 per cent of total calorie – include polysaccharides like whole grains – high fibre low GI foods preferred – low CHO improve insulin sensitivity – **2 marks**
- Protein – 15 – 20 % of calorie – supply essential amino acids for tissue repair – promotes satiety – 0.8 – 1g/kg body weight can be given – in type 1, 1-1.5 g/kg body weight – 0.6 g/kg body weight in renal condition – **2 marks**
- Fat – visible fat 20 – 25 g/day – saturated fat <7 % total calories – 15 – 25 % total calories, high in PUFA – include omega 3 fatty acids **- 1 mark**
- Dietary fibre – lower insulin requirement – increase tissue insulin sensitivity – decrease serum cholesterol and triglyceride values – aid in weight control and lower BP – soluble and insoluble fibre – **2 marks**
- Vitamins – supply antioxidant vitamin E and C – vitamin B12 – **½ mark**
- Minerals – chromium – zinc – sodium and potassium – **1 mark**

20. Explain the aetiology and assessment of obesity

- **Aetiology –** genetic factors – age and gender – eating habits – physical activity – stress – endocrine factor – trauma – prosperity and civilisation – drugs. **- 4 marks**
- **Assessment -** Body weight – BMI – Waist circumference- measurement of body fat – Ponderal index – waist to hip ratio – broka's index **- 6 marks**